

- Q.27 What are the causes and remedies of canal breaches?
(Three each)
- Q.28 Explain the method of developing a tube well.
- Q.29 Define confined and unconfined aquifers.
- Q.30 Describe the construction method of a zoned earth dam.
- Q.31 Explain the function of canal head works.(any five)
- Q.32 Describe with sketch the super passage cross-drainage work.
- Q.33 What is the purpose of canal escapes? (Any five)
- Q.34 List methods of river training and describe any two.
- Q.35 What are the causes and prevention methods of water logging? (Three each)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain in detail the methods of estimating rainfall and the factors affecting run-off.
- Q.37 What are the different types of tube wells? Explain their suitability, advantages, and method of determining yield.
- Q.38 a) Classify dams. Describe in detail gravity dams and their construction. (5-marks)
b) Explain various water harvesting techniques used in irrigation engineering.(5-marks)

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Roll No.

**4th Sem / Civil, Brick Tech., Constr. Mgmt,
Highway Engg.**

Subject:- Irrigation Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following crops requires the highest amount of water?
a) Wheat b) Rice
c) Maize d) Bajra
- Q.2 The area which can be irrigated with a unit discharge is known as
a) Delta b) Duty
c) Base period d) Crop period
- Q.3 The rain gauge which works without any manual effort is:
a) Symons gauge b) Tipping bucket
c) Float type d) Weighing type
- Q.4 Which method of irrigation is best suited for arid regions with less water?
a) Flow b) Lift
c) Drip d) Sprinkler

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Q.5 Gravity dams are mainly constructed using:

- a) Timber b) Steel
- c) Masonry or concrete
- d) Earth

Q.6 The canal lining is provided to:

- a) Improve water quality
- b) Increase seepage
- c) Reduce water loss
- d) Strengthen crops

Q.7 Cone of depression is related to:

- a) Canal irrigation b) Sprinkler system
- c) Well irrigation d) River training

Q.8 Cross drainage work where canal is above the drain is called:

- a) Aqueduct b) Super Passage
- c) Syphon d) Inlet

Q.9 The term “cut-off” in river training means:

- a) Raising of banks
- b) Canal lining
- c) Diversion of river path
- d) Construction of sluices

Q.10 Which structure is used to release surplus water from dam?

- a) Weir b) Barrage
- c) Spillway d) Regulator

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define irrigation.

Q.12 What is the base period of a crop?

Q.13 State any one advantage of lift irrigation.

Q.14 What is gross commanded area (GCA)?

Q.15 Write one use of canal outlets.

Q.16 What is the function of a strainer in tube well?

Q.17 Give one examples of an earth dam.

Q.18 What is the function of a canal head regulator?

Q.19 Define water logging.

Q.20 Mention one method of groundwater recharge.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Define duty, delta, and base period with their relationship.

Q.22 Explain the necessity of irrigation in India.(Any five)

Q.23 Differentiate between Kharif and Rabi crops with examples.

Q.24 What are the advantages and disadvantages of flow irrigation? (any five)

Q.25 Explain the layout and components of sprinkler irrigation system.

Q.26 Describe any two types of canal lining with sketches.

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